

S 2.7

P2. Solution:

$$AB = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$$

$$(AB)^T = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$$

$$A^T B^T = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$$

$$B^T A^T = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$$

$$\Rightarrow (AB)^T = B^T A^T \neq A^T B^T$$

P4. Solution:

To show $A^2 = 0$ is possible when $A \neq 0$.

$$A = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}, A^2 = 0$$

To show $A^T A = 0$ iff $A = 0$. Consider

A a 2x2 matrix such as

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \text{ then}$$

$$A^T A = \begin{bmatrix} a^2 + b^2 & ac + bd \\ ac + bd & c^2 + d^2 \end{bmatrix} = 0$$

$$\Leftrightarrow a^2 + b^2 = 0, c^2 + d^2 = 0$$

$$\Leftrightarrow a = b = c = d = 0, \text{ i.e., } A = 0$$

P6. Solution:

$$M^T = \begin{bmatrix} A^T & C^T \\ B^T & D^T \end{bmatrix}$$

when $A = A^T, D = D^T, B = C^T$,

M is symmetric.

P7. Solution:

(a) False.

(b) False

(c) True

(d) True

P18. Solution:

$$(a) A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ a_{31} & a_{32} & a_{33} & a_{34} & a_{35} \\ a_{41} & a_{42} & a_{43} & a_{44} & a_{45} \\ a_{51} & a_{52} & a_{53} & a_{54} & a_{55} \end{bmatrix}$$

$$1+2+3+4+5 = 15$$

General $n \times n$ symmetric matrix

$$1+2+3+\dots+n = \frac{n(n+1)}{2}$$

$$(b) A = L D L^T \rightarrow \begin{matrix} \downarrow & \searrow \\ & 5 \text{ diagonal} \\ & 15 \end{matrix}$$

$$1+2+3+4 = 10 > 15$$

$$(c) A^T = -A \Rightarrow a_{ii} = -a_{ii} \\ 2a_{ii} = 0 \Rightarrow a_{ii} = 0 \\ i = 1, 2, 3, 4, 5$$

$1+2+3+4 = 10$ entries can be chosen.

P22. Solution:

$$i) A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 2 & 3 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 2 & 3 & 4 \end{bmatrix} \xrightarrow{P_{12}} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 3 & 4 \end{bmatrix} \xrightarrow{e_{31} \leftarrow 2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 3 & 2 \end{bmatrix}$$

$$\xrightarrow{e_{32} \leftarrow 3} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix}$$

$$\Rightarrow PA = LU \text{ where}$$

$$P = P_{12} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix}$$

$$ii) A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 1 \\ 1 & 1 & 1 \end{bmatrix} \xrightarrow{P_{21} \leftrightarrow 1} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} \xrightarrow{P_{31} \leftrightarrow 1}$$

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix} \xrightarrow{P_{23}} \begin{bmatrix} 1 & 2 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$\Rightarrow PA = LU$ where

$$P = P_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, L = \begin{bmatrix} 1 & & \\ 1 & 1 & \\ 2 & 0 & 1 \end{bmatrix} (*)$$

$$U = \begin{bmatrix} 1 & 2 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Ex 3.1.

P4. solution:

$$\frac{1}{2}A = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \quad \text{vector } \vec{0} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$-A = \begin{bmatrix} -2 & 2 \\ -2 & 2 \end{bmatrix}$$

The smallest subspace of M containing A is the set of all matrices having the form αA for any scalar α .

P5. Solution:

(a) All matrices having the form $\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$, i.e., all diagonal matrices.

(b). Yes, it must contain I since

$$A - B = I$$

(c) All matrices having the form $\begin{bmatrix} 0 & 0 \\ a & 0 \end{bmatrix}$.

P9. solution:

$$(a) \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mid x_1 \geq 1, x_2 \geq 1 \right\}$$

$$(b) \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mid x_2 \geq 0, x_1 \geq x_2 \right\} \cup \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mid x_2 \leq 0, x_1 \leq x_2 \right\}$$

P11. Solution.

(a) All matrices $\begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix}$

(b) All matrices $\begin{bmatrix} a & a \\ 0 & 0 \end{bmatrix}$

(c) All matrices $\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$

P19. Solution:

$$(a) A = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$$

$$AX = x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 0 \end{bmatrix} = (x_1 + 2x_2) \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} c \\ 0 \end{bmatrix}$$

$C(A)$ is the set of all vectors $\begin{bmatrix} c \\ 0 \end{bmatrix}$

$$(b) B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

$$BX = x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} x_1 \\ 2x_2 \end{bmatrix}$$

$C(B)$ is the set of all vectors $\begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$

$$(c) C = \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix}$$

$$CX = x_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 0 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} c \\ 2c \end{bmatrix}$$

$C(C)$ is the set of all vectors $\begin{bmatrix} c \\ 2c \end{bmatrix}$